



Lecture 12

Effects

Modern Game Engine - Theory and Practice





Devil May Cry 5



Elden Ring



Final Fantasy XVI

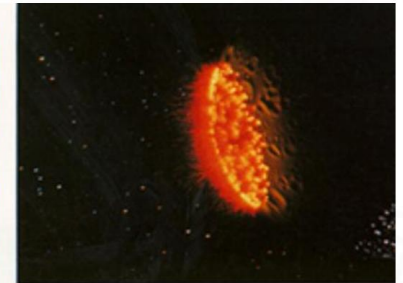


Sekiro: Shadows Die Twice

History of Particle System

“A particle system is a collection of many many minute particles that together represent a fuzzy object. Over a period of time, particles are generated into a system, move and change from within the system, and die from the system.”

—William Reeves, "Particle Systems—A Technique for Modeling a Class of Fuzzy Objects," ACM Transactions on Graphics 2:2 (April 1983), 92.



Star Trek II: The Wrath of Khan
(first introduced particle system, 1982)

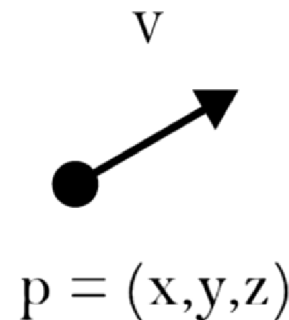


Particle

A particle in game is usually a sprite or 3D model, with the following attributes:

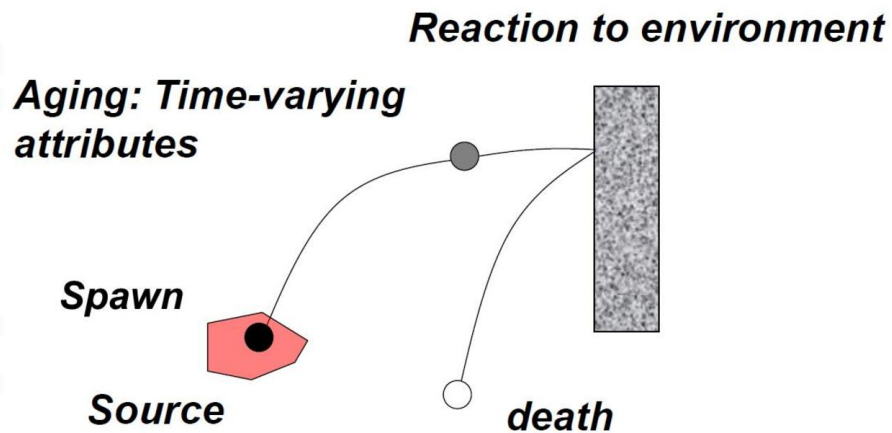
- Position
- Velocity
- Size
- Color
- Lifetime

.....





Particle's Life Cycle

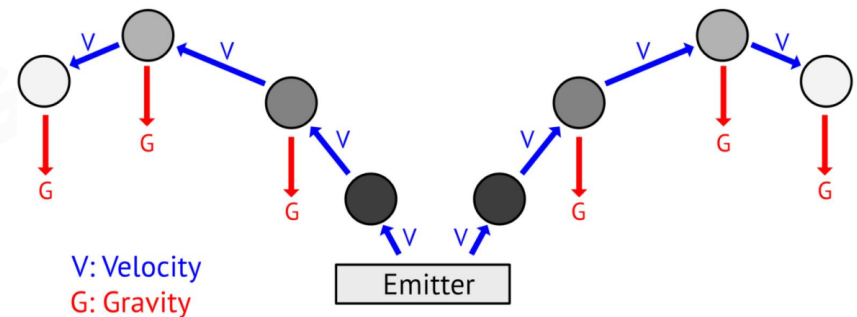




Particle Emitter

Particle Emitter is used to define the particles simulation

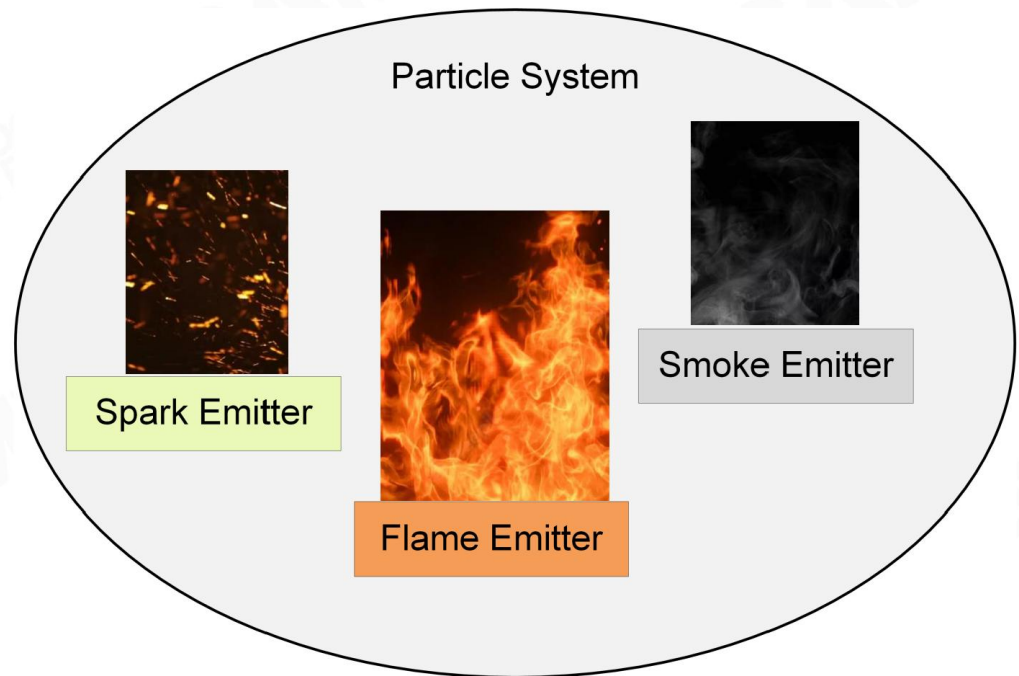
- Specify the spawn rules
- Specify simulation logic
- Describe how to render particles





Particle System

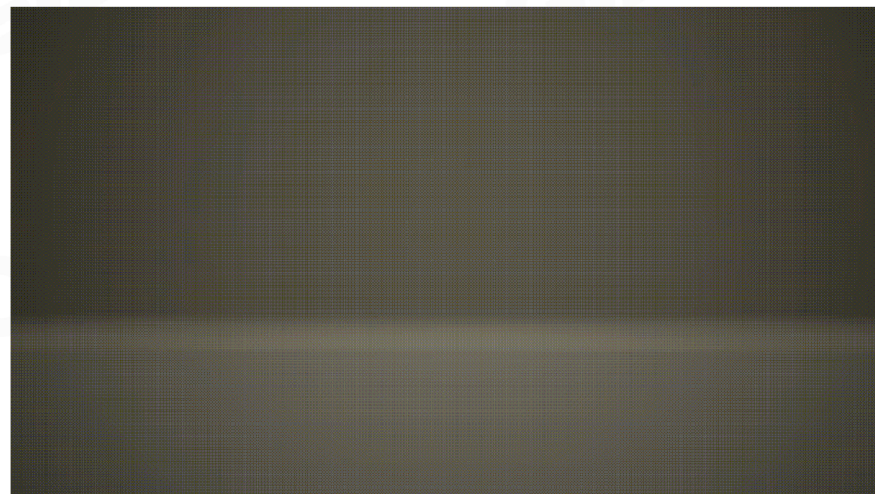
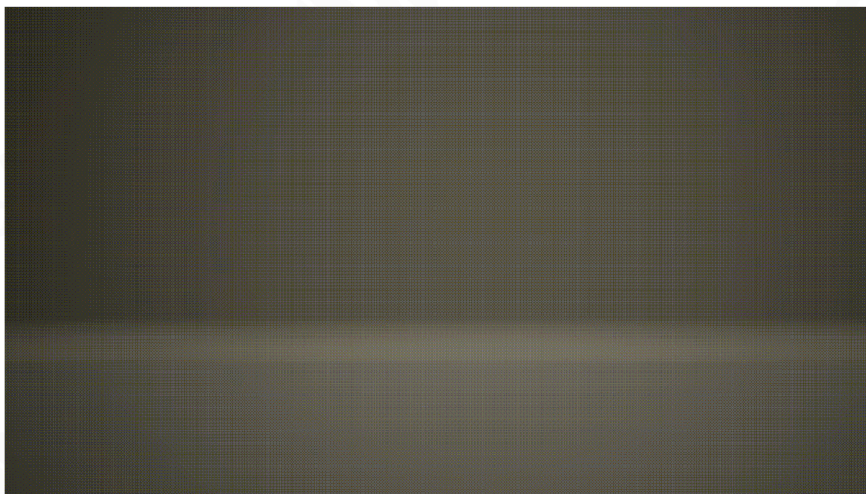
A particle system is a collection of individual emitters





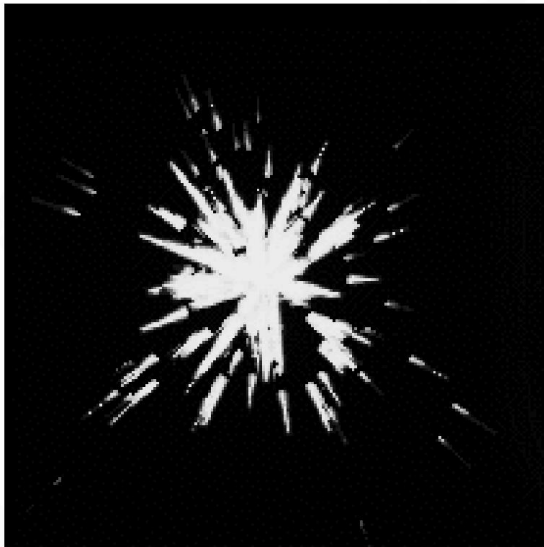
Particle System

A simple case, just combine different emitters

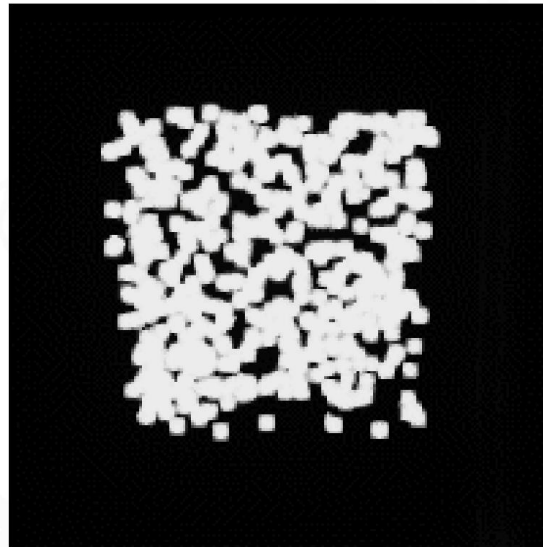




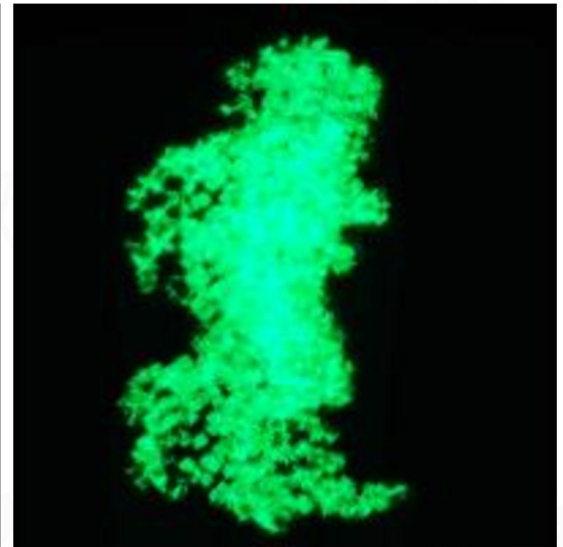
Particle Spawn Position



Single position spawn



Spawn based on area



Spawn based on mesh



Particle Spawn Mode

Continuous

- variable spawn rate per frame
- time, distance based, etc.

Burst

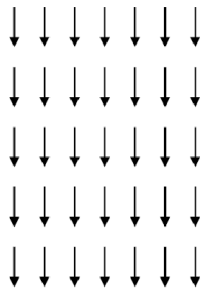
- all particle spawn and simulated at once.





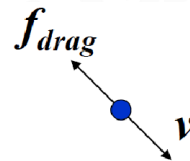
Simulate (1/5)

Common forces



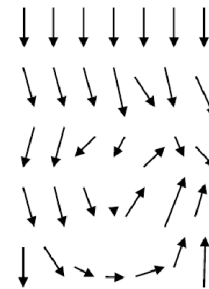
Gravity

$$f = mg$$



Viscous Drag

$$f = -k_d v$$



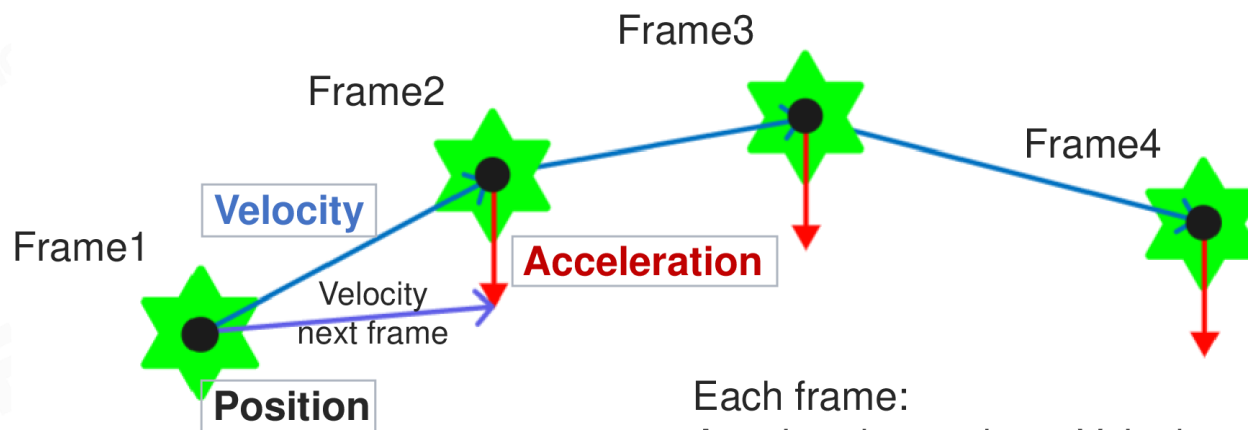
Wind Fields

$$f = k v_{wind}$$



Simulate (2/5)

Simulate controls how the particles change over time

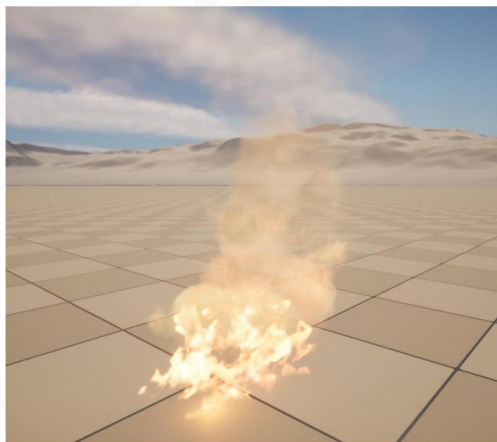


Each frame:
Acceleration updates Velocity
Velocity updates position

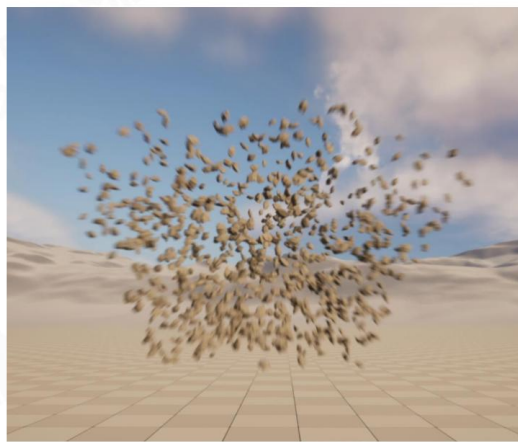


Particle Type

- Billboard Particle
- Mesh Particle
- Ribbon Particle



Billboard Particle



Mesh Particle



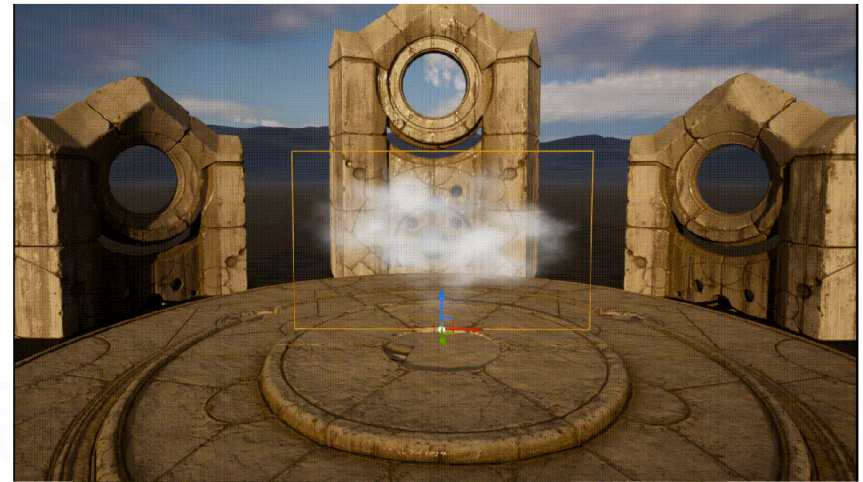
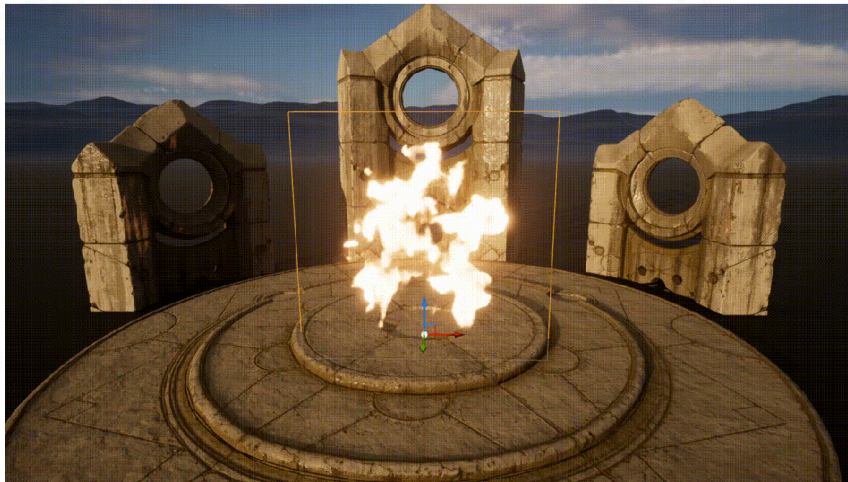
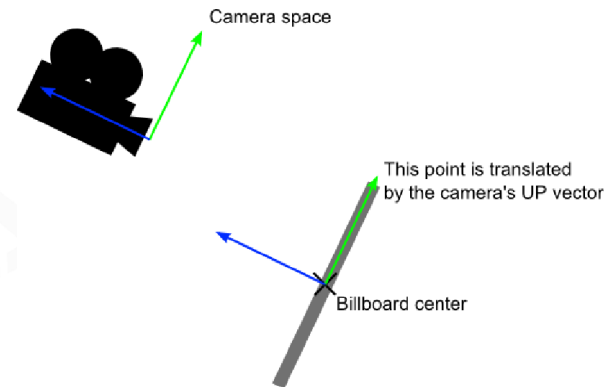
Ribbon Particle



Billboard Particle

Each particle is a sprite

- Appears to be 3D
- Always face the camera

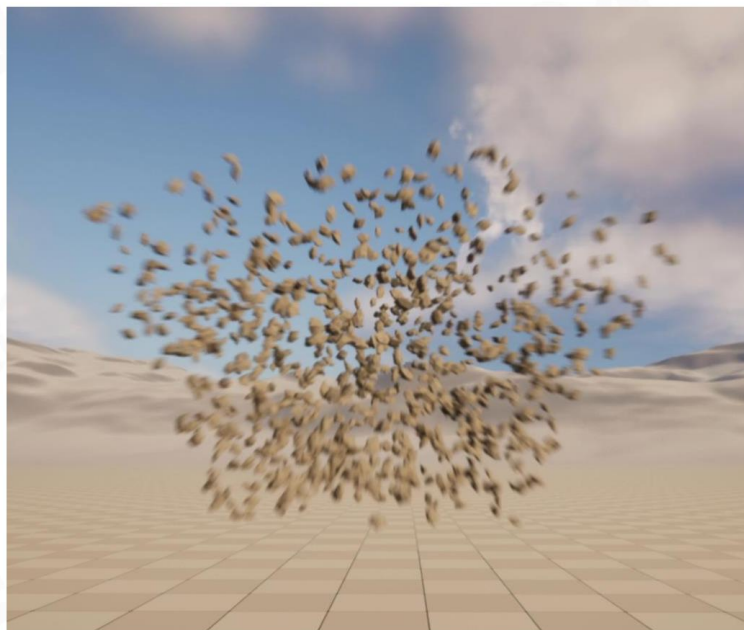




Mesh Particle

Each particle is a 3D model.

- Rocks coming from a explosion

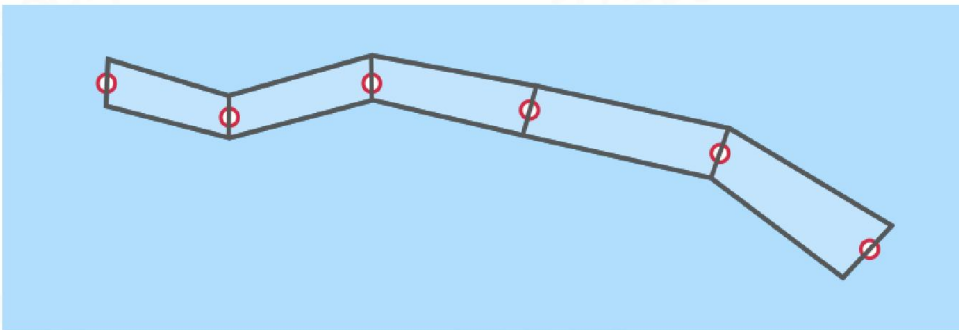




Ribbon Particle

A strip is created by connecting the particles and rendering quads between the adjacent particles.

- particles (represented as red dots)



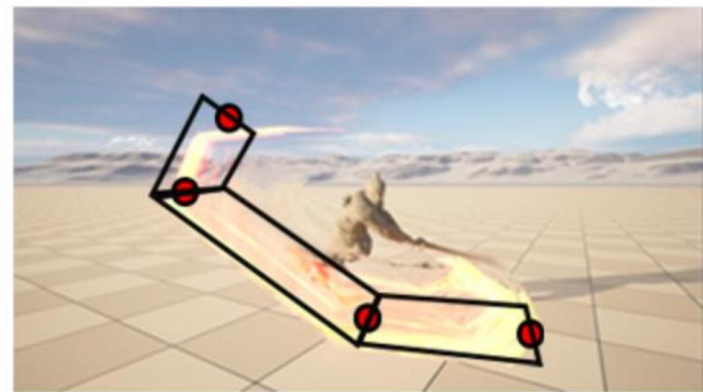
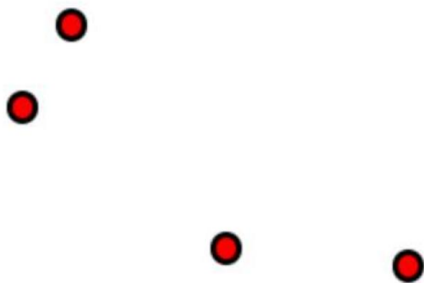
Ribbon Particle:Slash



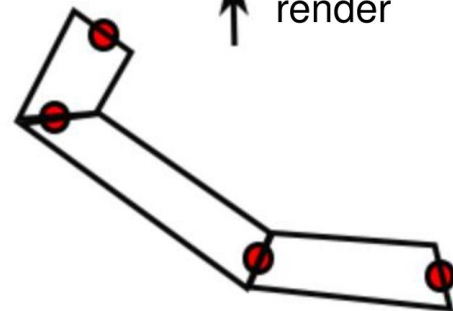
Ribbon Particle Case(1/3)



spawn



render

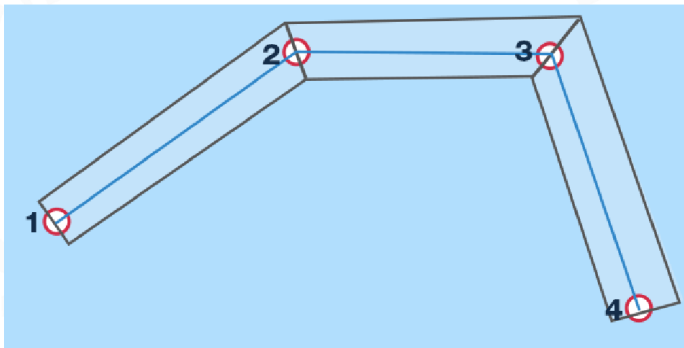




Ribbon Particle Case(2/3)

No smoothing shape

- with sharp angles



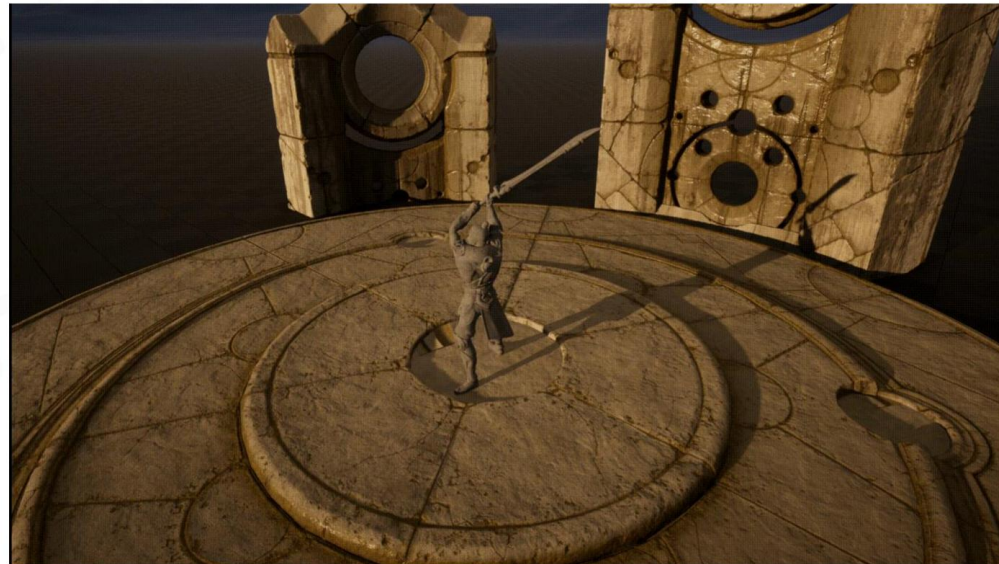
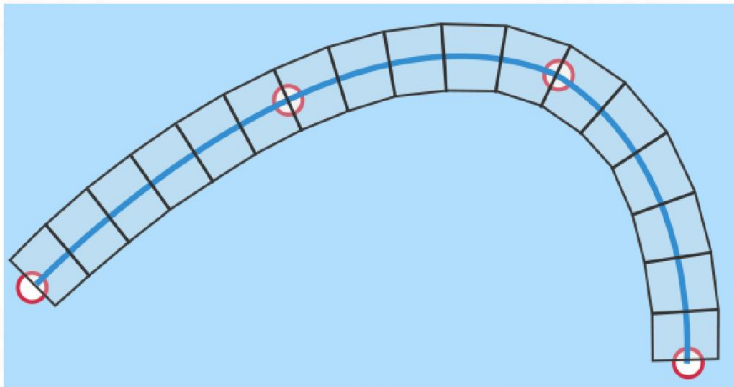
Slash without smoothing



Ribbon Particle Case(3/3)

Smoothing with Centripetal Catmull-Rom interpolation

- add extra segments between particles
- can set the number of segments
- requires more CPU



smoothing with Centripetal Catmull-Rom



Particle System Rendering



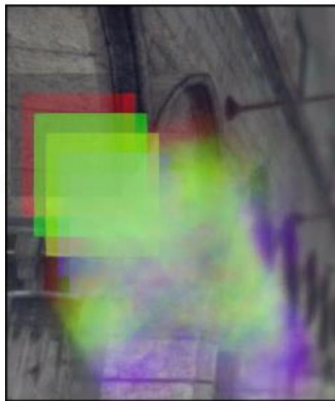
Alpha Blending Order

$$C_f = a_1 C_1 + (1 - a_1) C_0$$

Final color

Source color

Destination color



Blend result of unsorted elements



Blend result of sorted elements



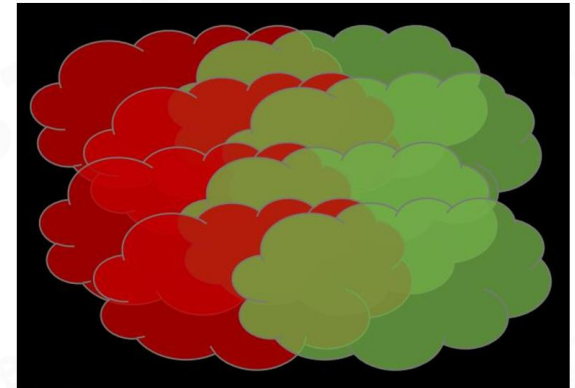
Particle Sort

Sorting mode

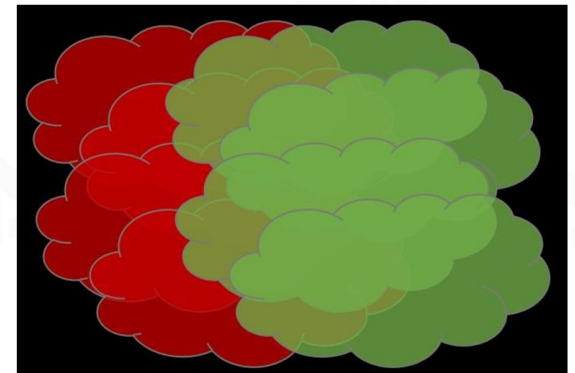
- Global
 - Accurate, but large performance consumption
- Hierarchy: Per system -> Per emitter -> Within emitter

Sort rules

- Between particles: based on particle distance with camera
- Between systems or emitters: bounding box



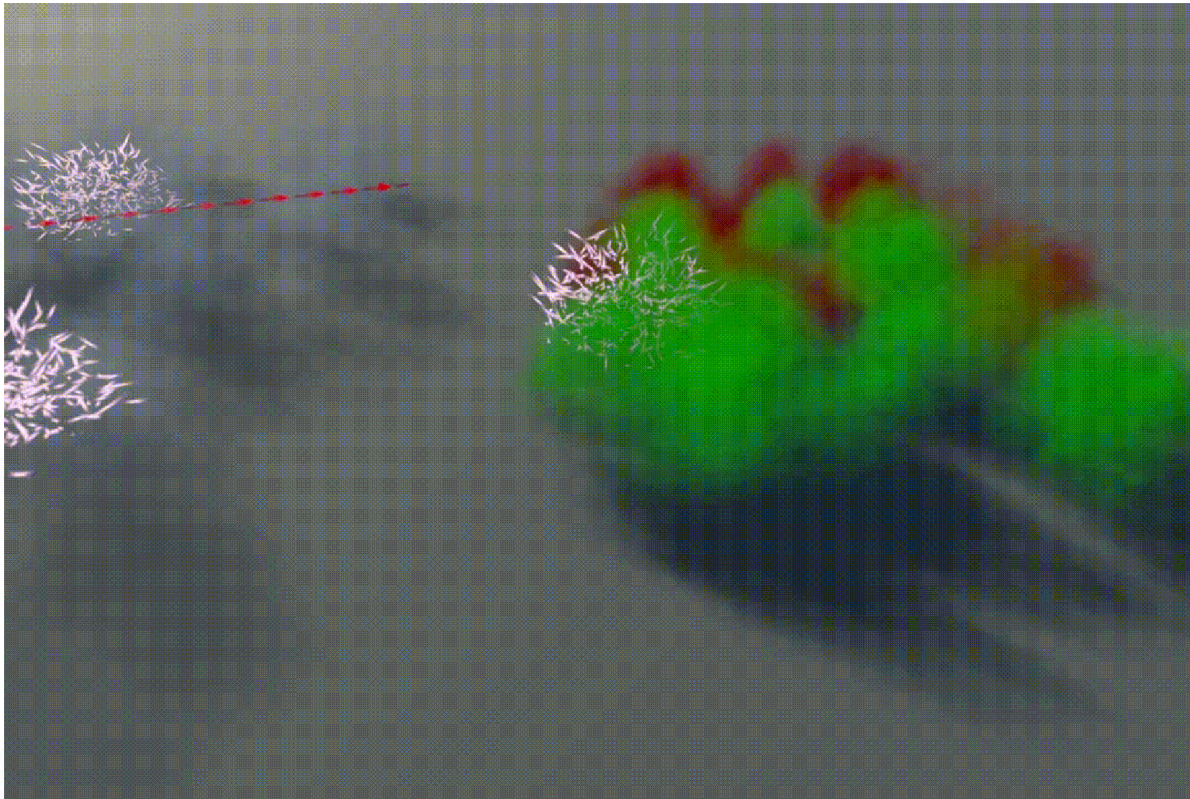
Global sort



Per emitter sort

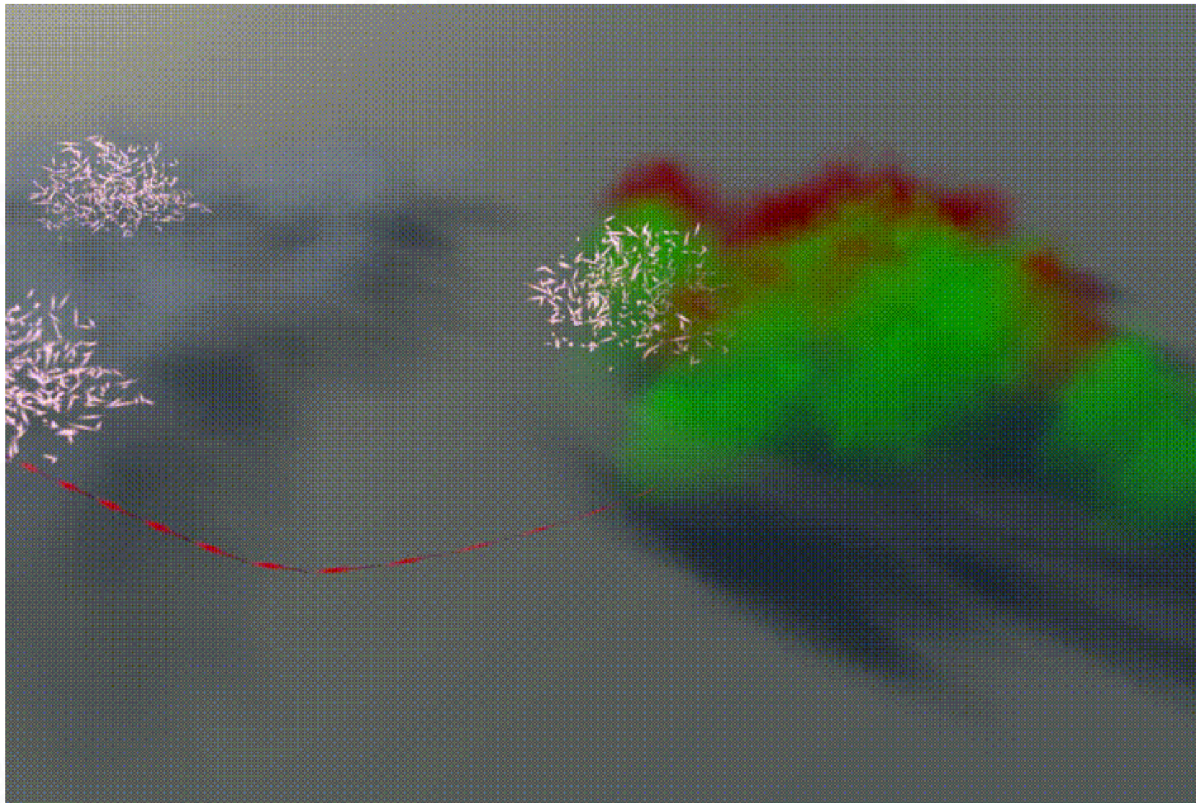


Per Emitter Sort





Global Sort





GPU Particle



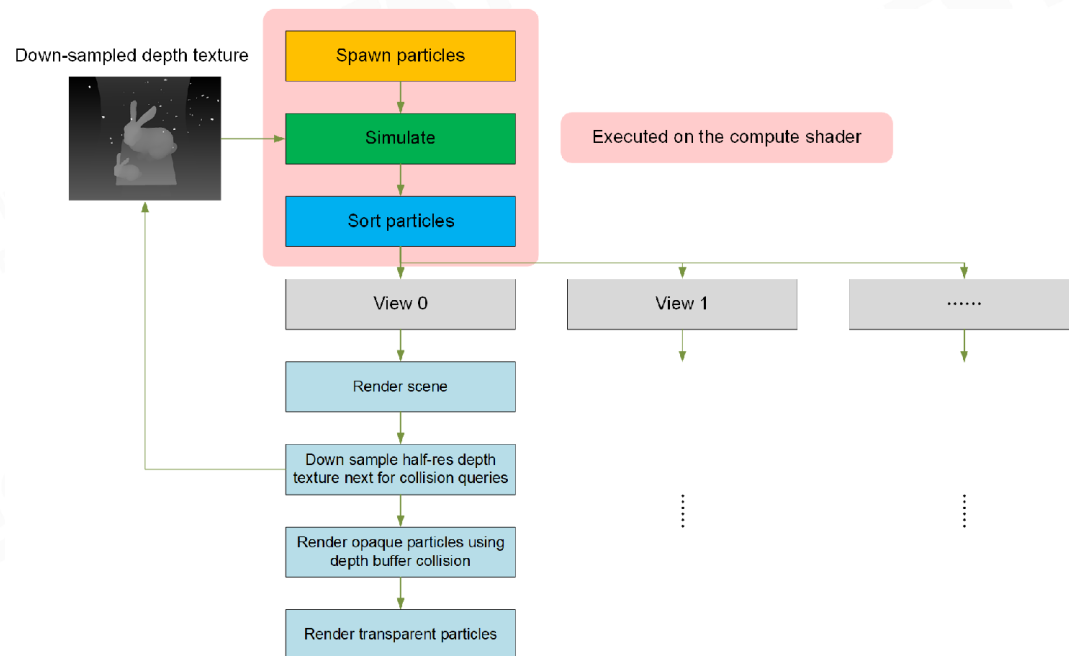
Processing Particles on GPU

Why use GPU?

- Highly parallel workload, suitable for simulation of large numbers of particles
- Free your CPU to do game code
- Easy to access depth buffer to do collision

GPU Particles - Frame Overview

Compute shader provides high-speed general purpose computing and takes advantage of the large numbers of parallel processors on the GPU

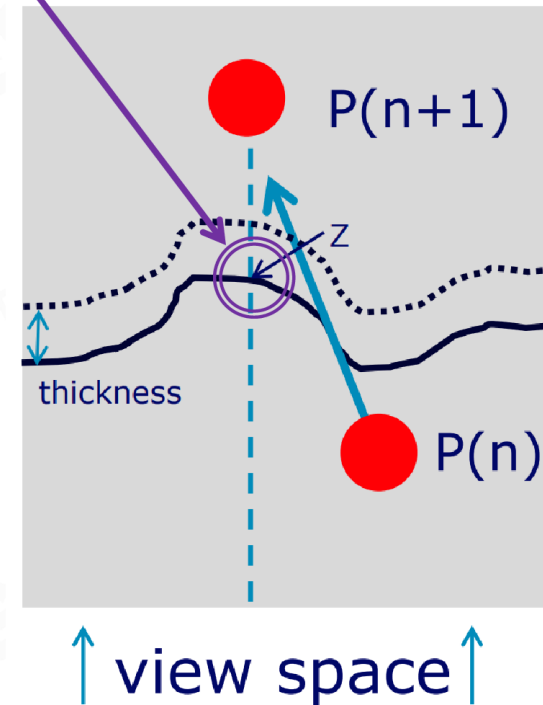




Depth Buffer Collision

1. Reproject particle position to previous frame screen space texture coordinate
2. Read depth value from previous frame depth texture
3. Check if particle is colliding with the depth buffer, but not completely behind it (where thickness value is used)
4. If collision is happened, calculate surface normal and bounce off the particle

Collision position





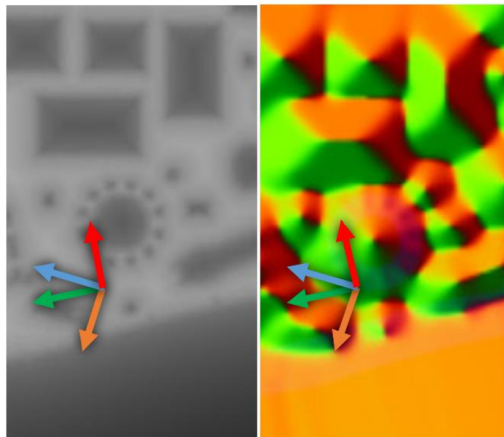
Advanced Particles



Crowd Simulation

Crowds - Runtime Behavior

- Design target locations to guide the movement of crowds
- The desire moving towards the target location, pushing away from blocking geometry, all become forces to influence the movement of crowds (if close enough, the camera also acts as a force)





Utilizing Particle System in Games

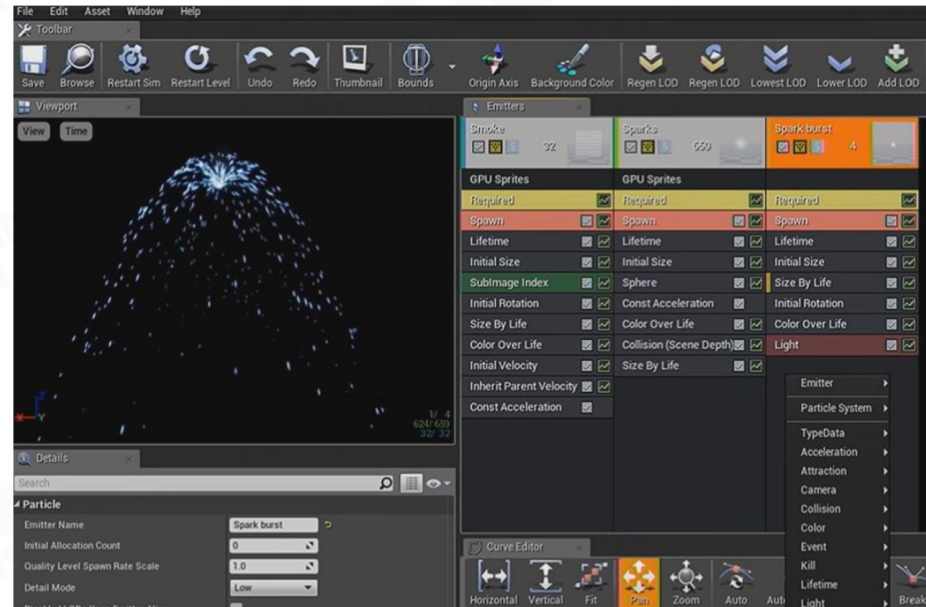
Design Philosophy - Preset Stack-Style Modules

Pros

- Fast to add behaviors as stacked modules
- Non technical artists have lots of control via a suite of typical behaviors
- Easy to understand at a glance

Cons

- Fixed functions, new feature requires new code
- Code-based, divergence in game team code
- Fixed particle data, data sharing is mostly impossible

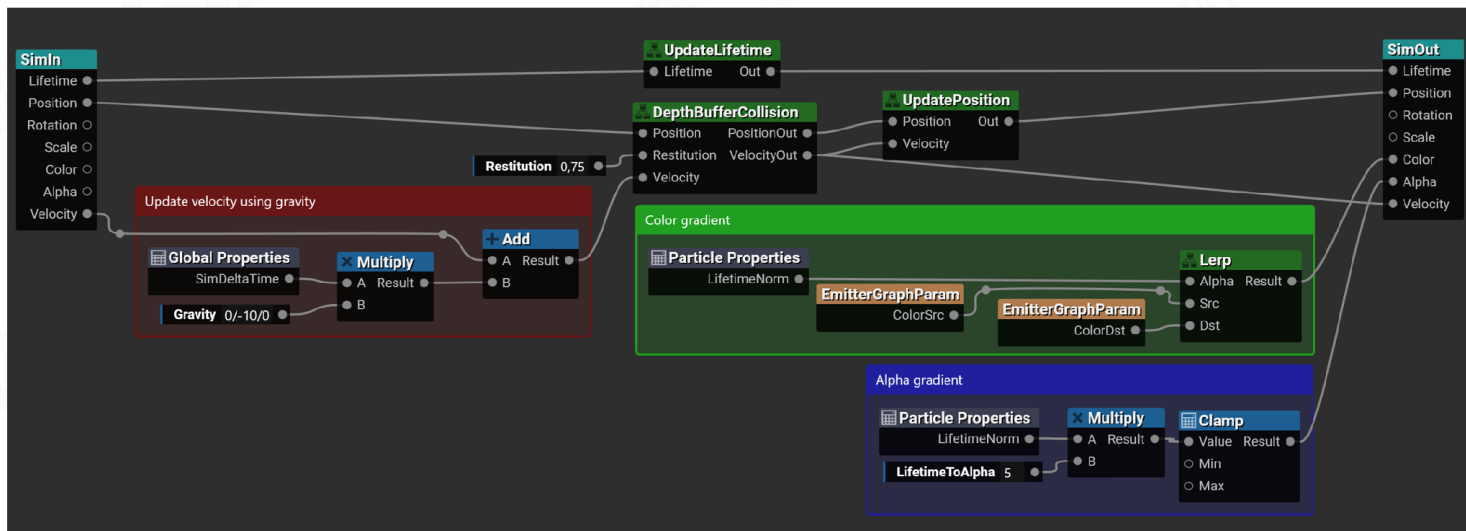


Cascade Particle System in Unreal Engine



Design Philosophy - Graph-Based Design

- Parameterizable and shareable graphs asset
- Less code divergence
- Provide modular tools instead of hardcoded features





Hybrid Design

- Graphs give total control
- Stacks provide modular behavior and glance readability

Modules	Emitters	Systems
• Encapsulate behaviors (seen as "write functions") • Stack with each other	• Containers for modules • Single purpose, reusable	• Holders of multiple emitters into one "Effect"
Graph	Stack	Stack

Unreal's Niagara System Design



Sound System



Audio

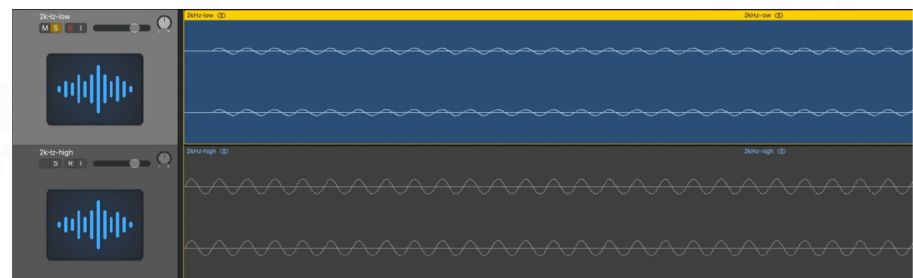
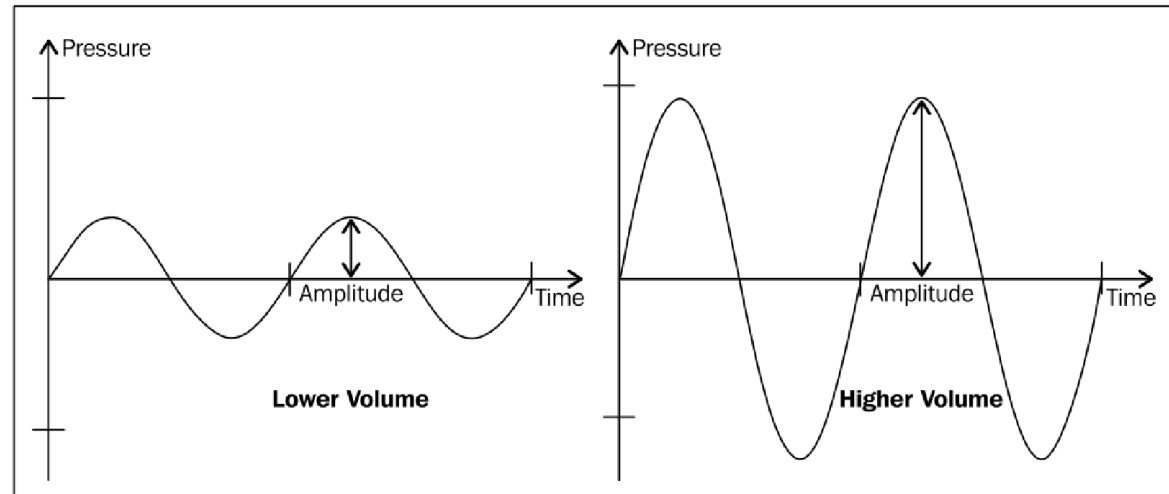
- Entertains the player
- Enhances the realism
- Establishes atmosphere





Volume

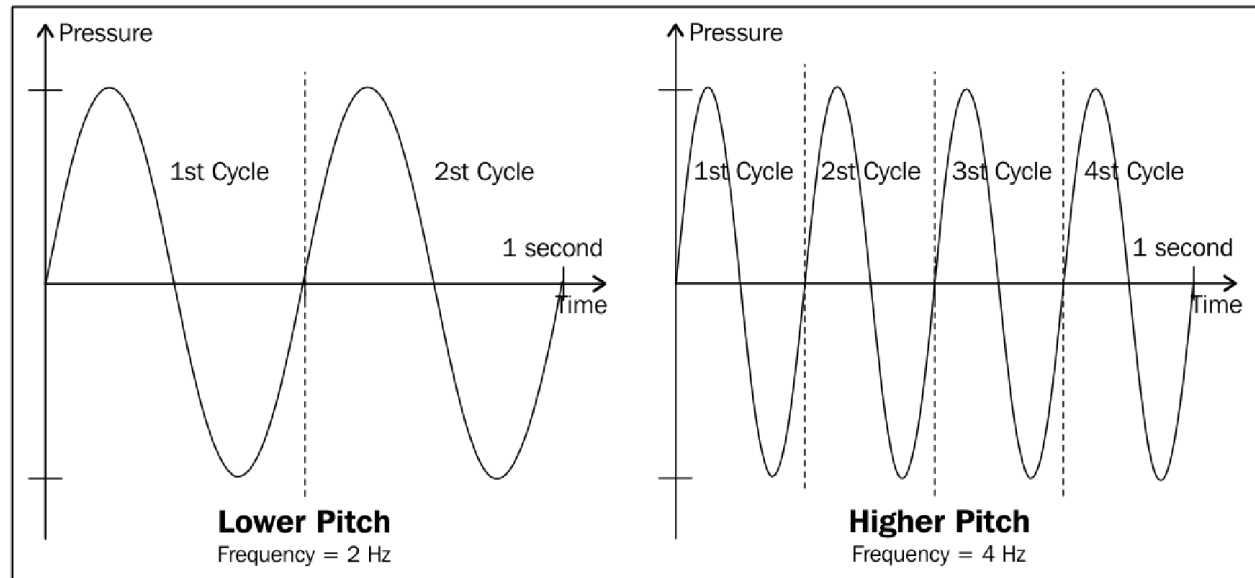
The amplitude of the sound wave





Pitch

- Determines how high or low the sound is
- Depends on the frequency of the sound wave



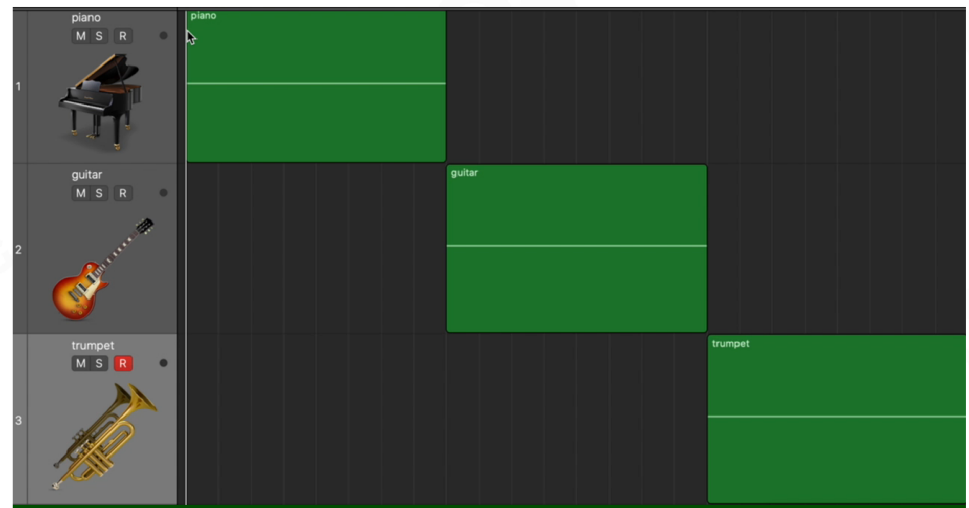


Timbre

Combinations of overtones or harmonics

- frequencies
- relative intensities

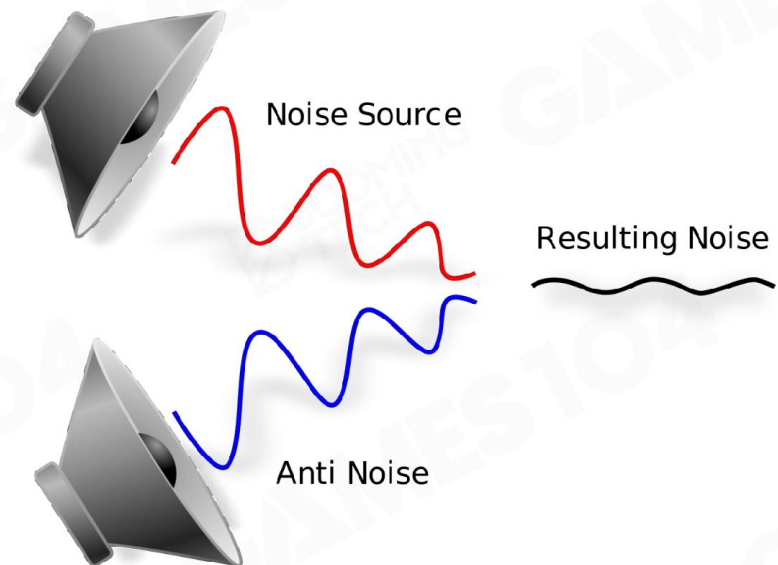
음색





Phase and Noise Cancelling

Same frequency, amplitude, but in different phase

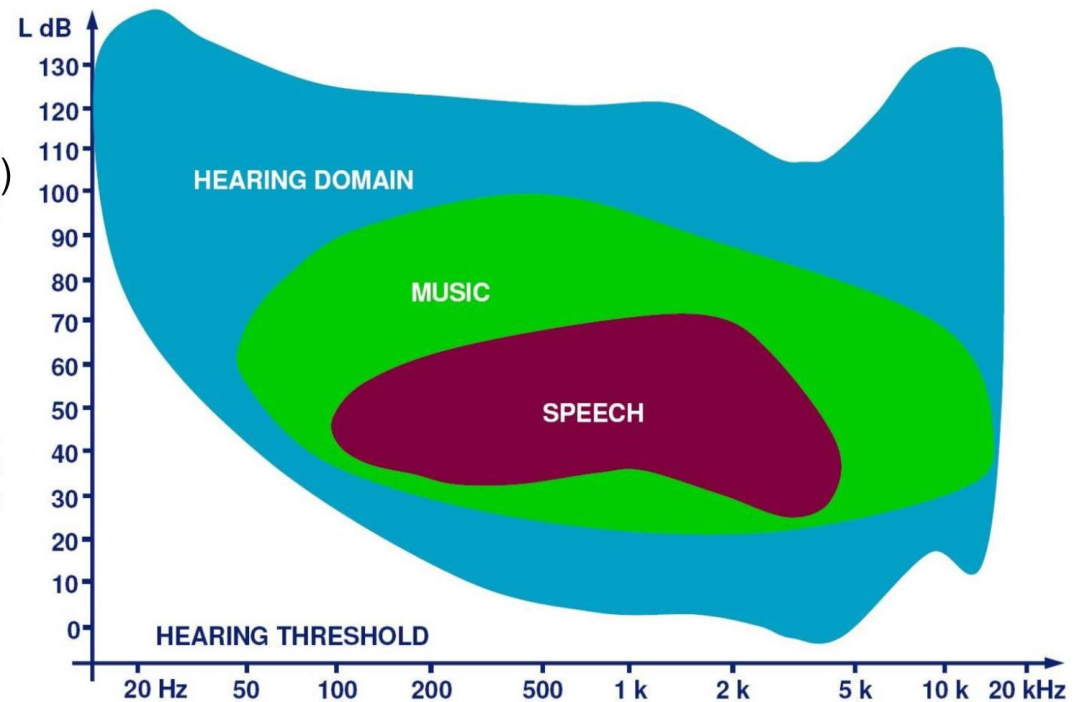




Human Hearing Characteristic

The audible sound of human ear

- frequency range: 20-20KHz
- sound pressure level range (0-130db)



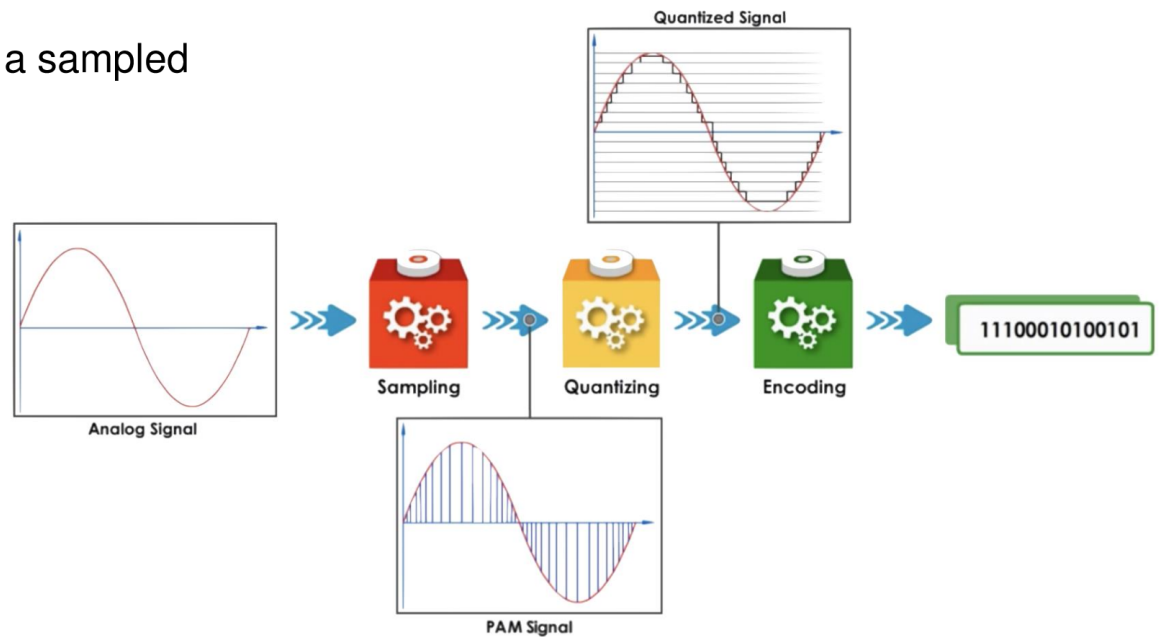


Digital Sound

Pulse-code Modulation (PCM)

Standard method for encoding a sampled analog sound signal

- Sampling
- Quantizing
- Encoding

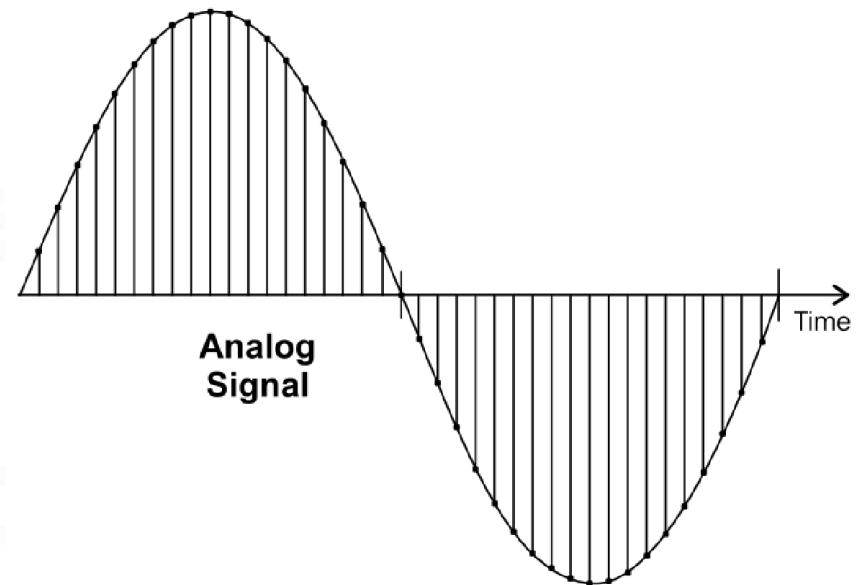




Sampling

Sample Frequency: Samples per second(Hz)

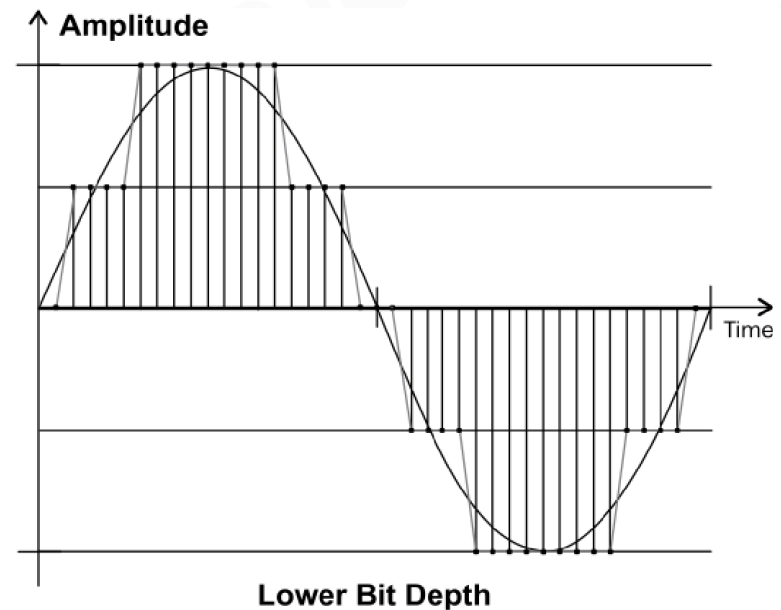
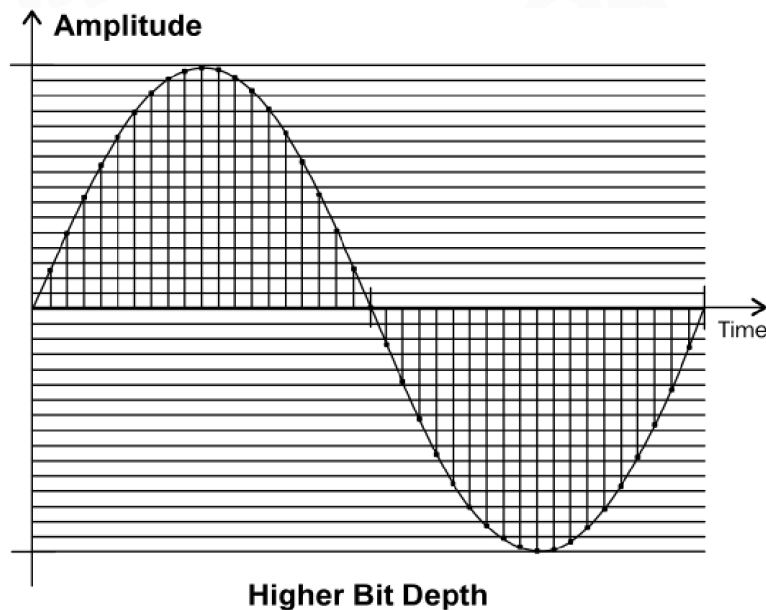
Nyquist–Shannon Sampling Theorem: *The minimum sampling frequency of a signal that it will not distort its underlying information, should be double the frequency of its highest frequency component*





Quantizing

bit-depth: bit depth is the number of bits of information in each sample





Audio Format

Format	Quality	Storage	Multi-channel	Patent
WAV(uncompressed)	★★★	★	★★★	★★★
FLAC(lossless)	★★★	★★	★★★	★★★
MP3(lossy)	★	★★★★	★	★
OGG(lossy)	★	★★★★	★★★★	★★★

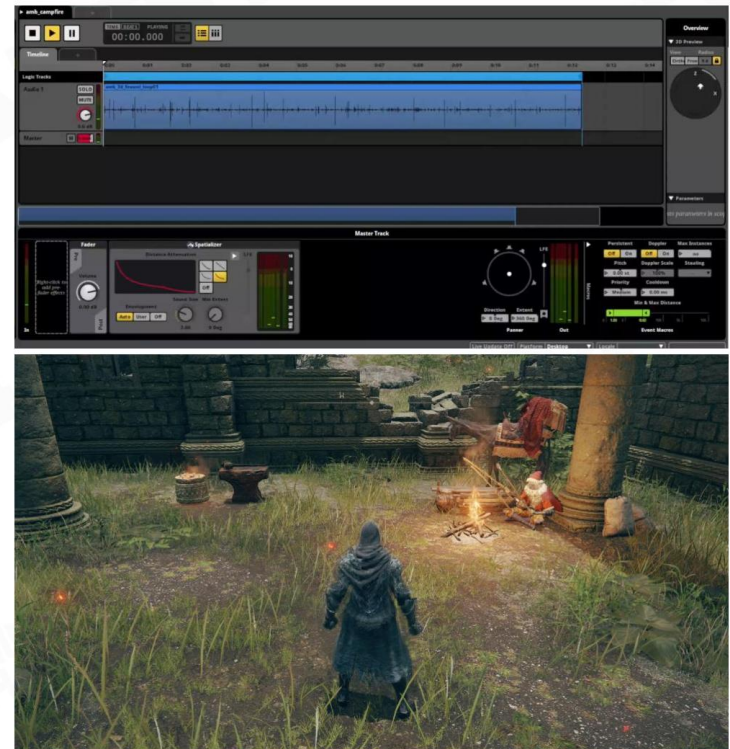


3D Audio Rendering



3D Sound Sources

- a mono-phonetic audio signal
- emanating from a specific position





Listener

A "virtual microphone"

- position
- velocity
- orientation



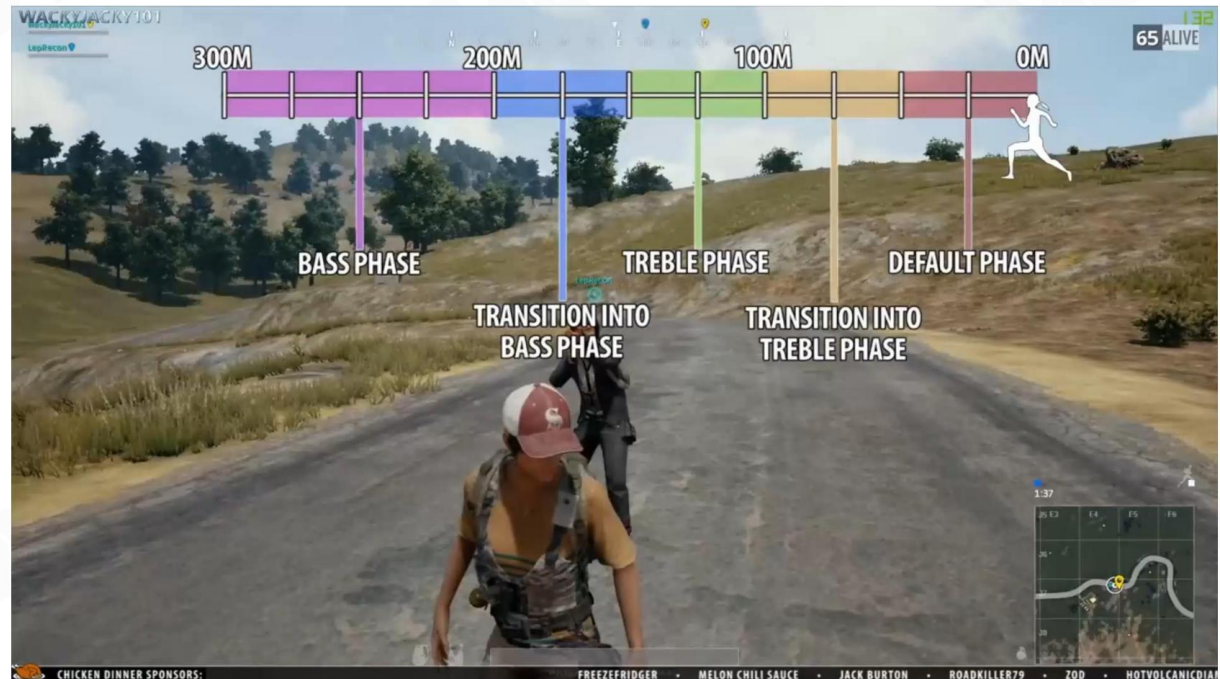


Attenuation

Volume will attenuate as the listener moves away from it

In real world, the sound pressure (p) of a spherical sound wave decreases as $1/r$ from the centre of the sphere:

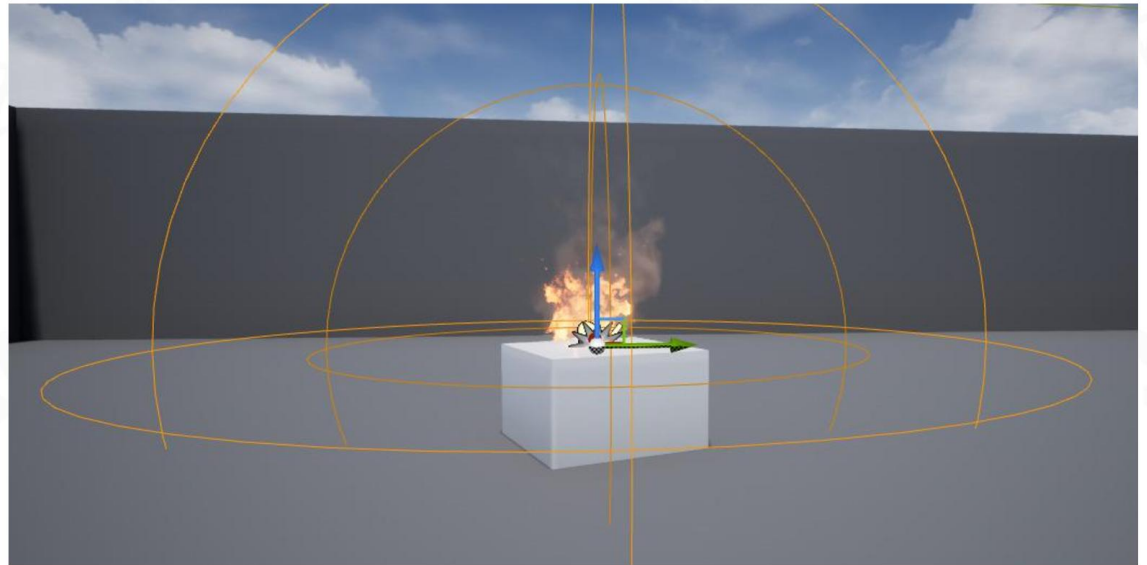
$$p(r) \propto \frac{1}{r}$$





Attenuation Shape - Sphere

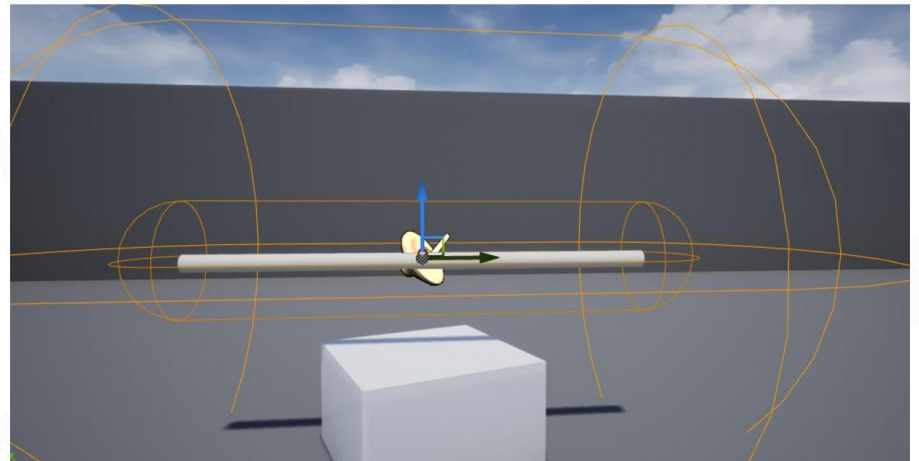
Useful for most spot sounds as it models how sound propagates in the real world





Attenuation Shape - Capsule

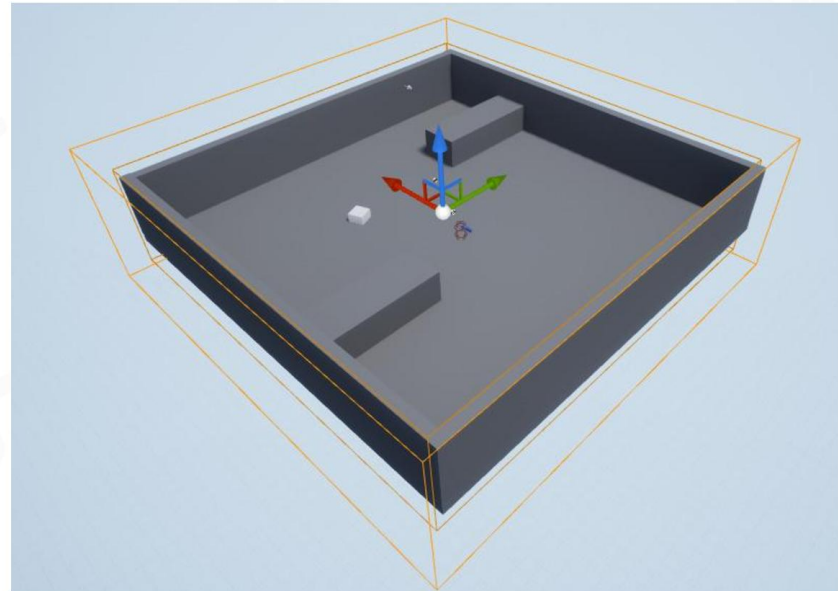
Useful for things like water pipes, where the sound doesn't want to appear to come from a single, specific point in space — the sound of gurgling water would follow the length of the pipe





Attenuation Shape - Box

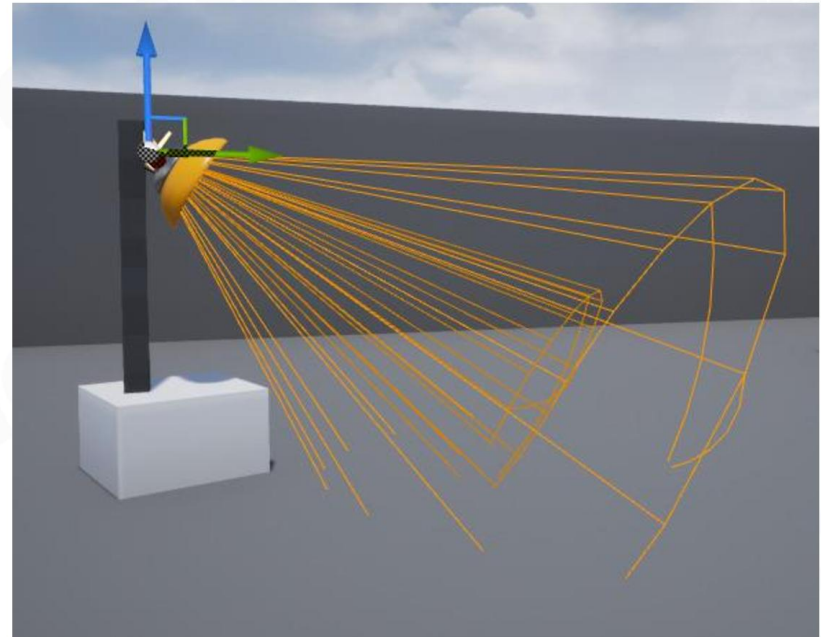
useful for things like room tones/ambience
as you can define the shape of the box to
match that of the room





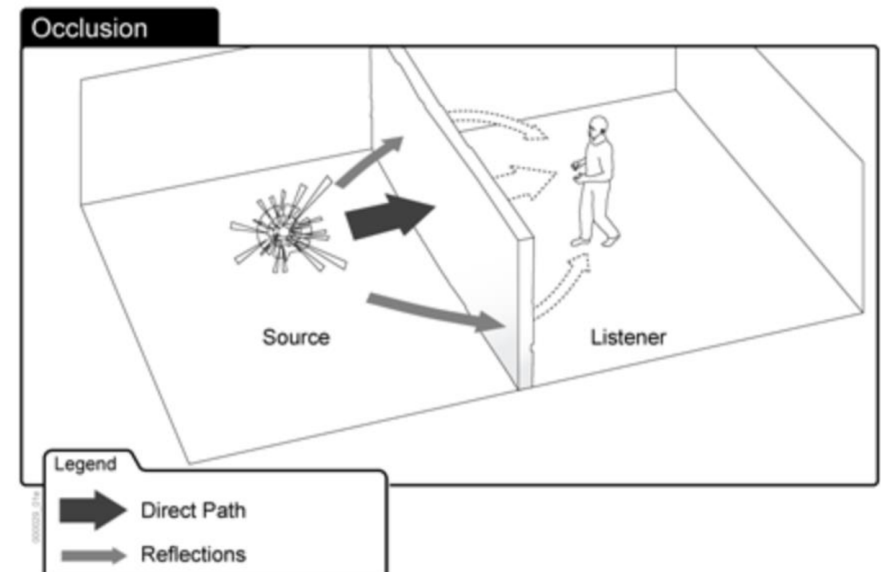
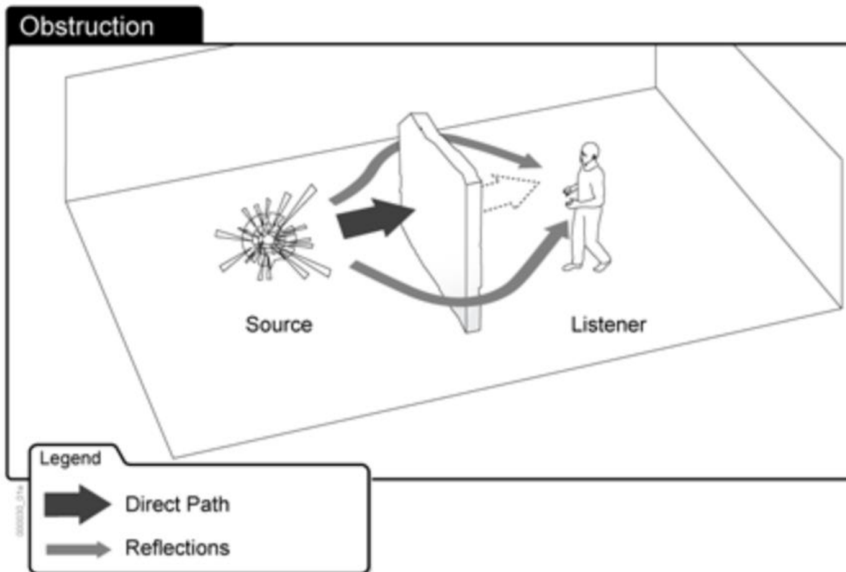
Attenuation Shape - Cone

Useful in situations when you want a directional attenuation pattern — for example, public address speakers





Obstruction and Occlusion (1/2)

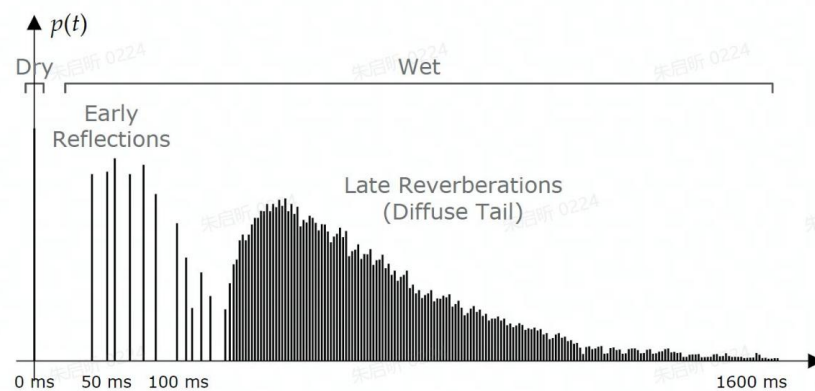
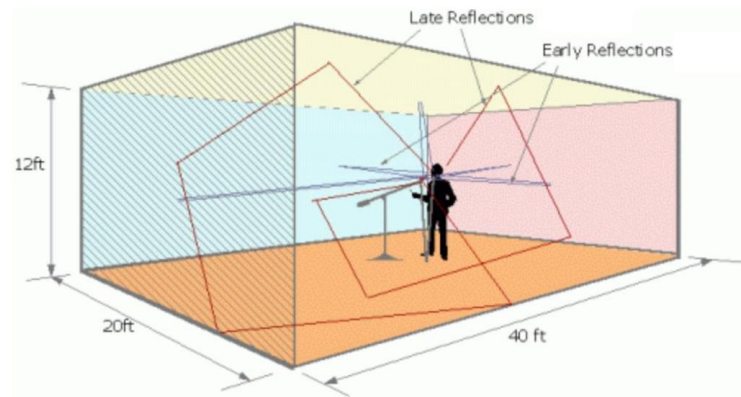




Reverb (1/3)

In any environment containing sound-reflective surfaces, a listener generally receives three kinds of sound waves from a sound source

- Direct (dry)
- Early reflections (echo)
- Late reverberations (tail)





Reverb (2/3)

Reverberation Time: Measure of how fast the sound dies away in a given room. The size of the room and the choice of materials determine the reverberation time

Absorption: Absorption coefficient of a material and material count determine the absorption

$$A = S \times \alpha (m^2 sab)$$

$$T = 0.16 \times \frac{V}{A} (s)$$

α : absorption coefficient

S: square

A: equivalent absorption area

V: volume in the room

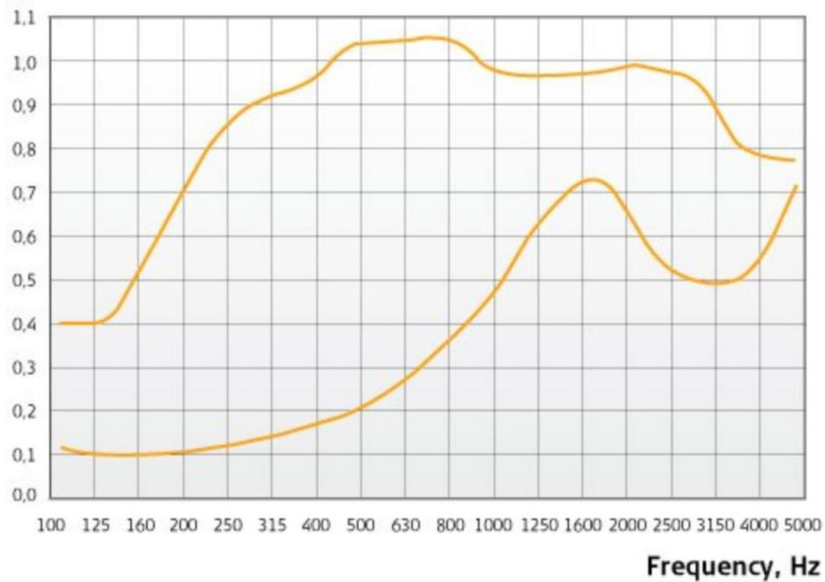
T: reverberation time

0.16: proportionality factor, the time (in seconds) it takes for the initial sound pressure level to be reduced by 60 dB



Reverb (3/3)

Sound absorption coefficient α



Materials/Locations	Sound Absorption Coefficients over 1/1 Octave Bands					
	125 Hz	250 Hz	500 Hz	1 kHz	2 kHz	4 kHz
Ornamented stone piers, arches columns, and column heads	0.05	0.05	0.06	0.09	0.11	0.11
Current carpet	0.02	0.12	0.25	0.38	0.55	0.58
Carpet with prayers	0.18	0.22	0.41	0.58	0.69	0.72
Historical plasters tested in 30% humidity	0.10	0.17	0.23	0.29	0.32	0.32
Current plasters on brick	0.13	0.09	0.07	0.05	0.03	0.04
Large pane of glass	0.18	0.06	0.04	0.03	0.02	0.02
Marble slabs	0.01	0.01	0.01	0.01	0.02	0.02
Wooden doors and furniture	0.10	0.07	0.05	0.04	0.04	0.04
Gypsum moldings for muqarnas and similar decorations	0.29	0.10	0.05	0.04	0.07	0.09
Copper decorative elements	0.12	0.08	0.02	0.01	0.01	0.01

Sound absorption coefficient and scattering data of applied materials in acoustical simulations



Sound in Motion: The Doppler Effect

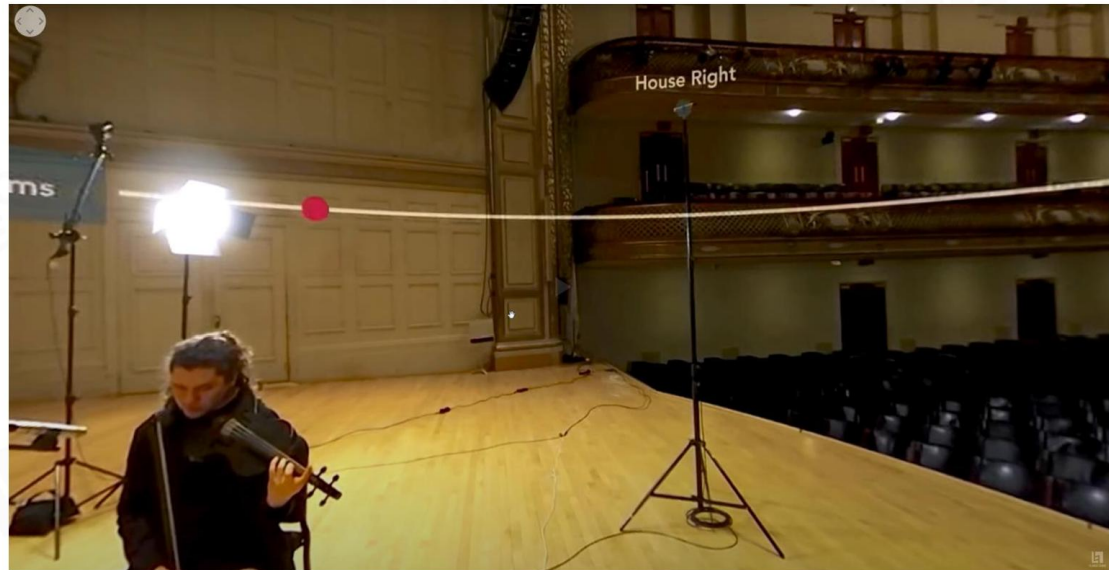
The change in frequency of a wave in relation to an observer who is moving relative to the wave source





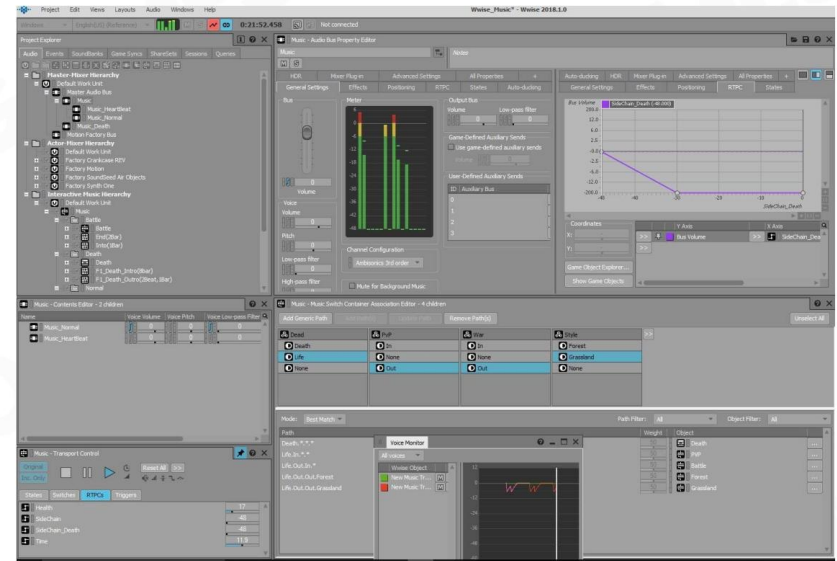
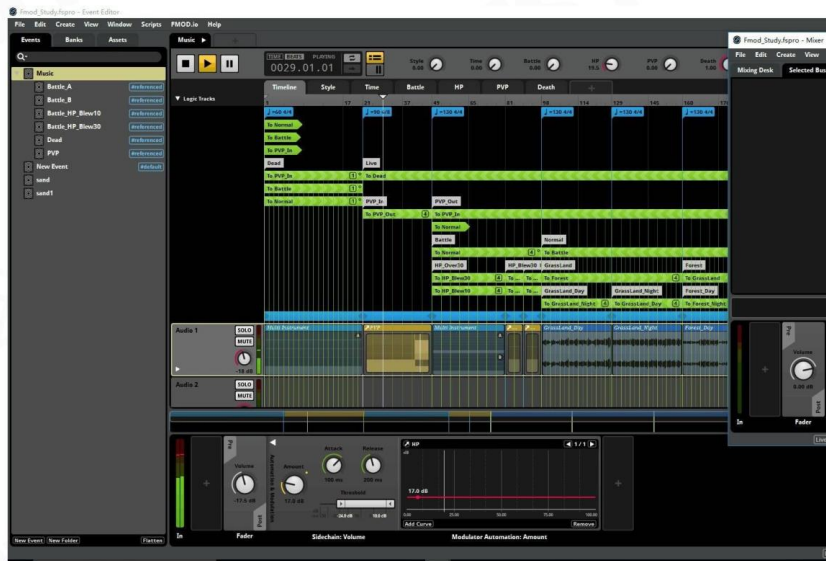
Spatialization - Soundfield

- **Full-sphere** surround sound
- Also known as ambisonics
- Used in 360 videos and VR





Common Middlewares





How does Audio Middleware Work?

